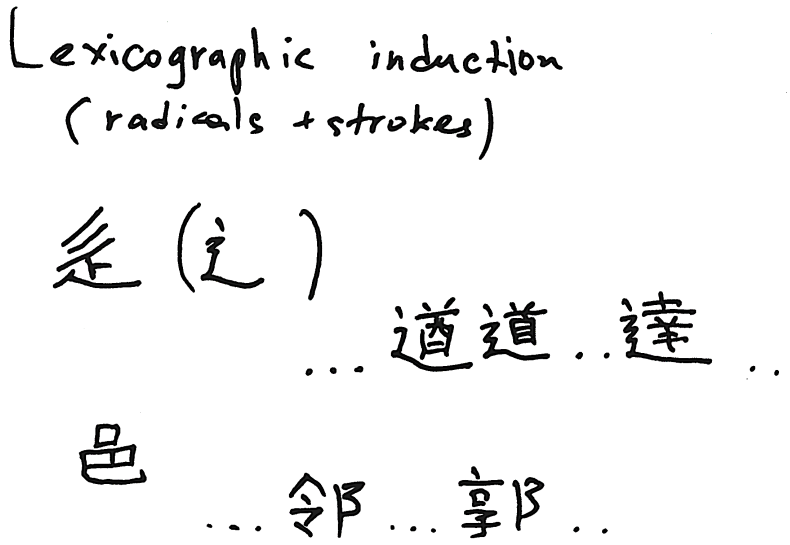
Basic concepts in mathematics: Diary

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https://kilin-math.github.io/assets/teaching/diary2.pdf
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0 Preface

From a cup of coffee, a spoonful of coffee is poured into a cup of milk. Then a spoonful of the resulting mixture is poured back into the cup of coffee. Which is greater: the amount of milk in the cup of coffee or the amount of coffee in the cup of milk?

This course is intended for first-year students in mathematics and computer science; they have to learn how to read and write proofs.

It is not obvious at all why we need to prove something (professors demand it, but why?). The truth is that **by proving we understand things better and discover new beauties** (e.g., the formula $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + .. + n)^2$).

So, to reconcile the concept of proof with students, it is better to introduce proofs in questions whose answers are not obvious and are debatable. Examples logic puzzles, games (who has a winning strategy?), impossibility proofs (e.g., tiling a chessboard missing opposite corners with dominoes), and induction. These topics occupy the first few lectures.

How do you learn to prove? Let us use the metaphor "Mathematics is a language". When you learn a foreign language, you have different activities: listening, speaking, reading, writing. The same is true for mathematics. To study it, students should spend time reading and writing mathematics, listening to lectures, thinking about problems, and discussing ideas.

Also, mathematicians love mathematics because proofs are beautiful!

Solution to the coffee-milk problem:

Let's try to guess the answer. To do that, consider an extreme case (this is the first idea). Suppose there is just one spoonful of liquid in each cup. Then after pouring the coffee into the milk, we take the entire mixture back. The mixture will be uniform, so the amount of coffee and milk will be equal. Will it always be equal?

Since one spoonful was poured "there and back," the total volume of liquid in each cup did not change (this is the second idea).

Therefore (the third idea), the amount of coffee lost equals the amount of milk gained.

The volumes of coffee and milk in the cups can be different, you can pour the spoon back and forth ten times, you can even stir the mixture poorly — it doesn't matter: the amount of milk in the coffee will always equal the amount of coffee in the milk!

0.1 :: on collaboration and the use of AI

Collaboration. You are encouraged to discuss problems with classmates: compare approaches, explain ideas to one another, and ask for critique. Explaining your reasoning often reveals subtleties and gaps that you can then address. However, be sure that you can write up the solution independently, in your own words, without assistance of friends or AI.

Attempt first. Before seeking help from classmates or AI tools, make a genuine attempt on each problem: write down an outline, partial calculations, or a strategy you tried (even if it failed). Learning to prove requires practice—like training for a sport—so expect to try, err, and revise.

AI: permitted uses. You may use AI tools to: (i) spot logical gaps or unclear steps in a solution you have already written, (ii) improve clarity and writing style, (iii) receive high-level hints about relevant definitions or theorems. When you do so, begin with your own draft (typed or a photo) and ask for feedback on that draft.

AI: not permitted uses (unless explicitly allowed). Do not ask AI to produce full solutions for graded assignments or to translate someone else's solution into "your own words." Do not paste problem statements and accept verbatim answers. Use AI as a reviewer, not as an author. If you are unsure whether a use is allowed, ask the instructor.

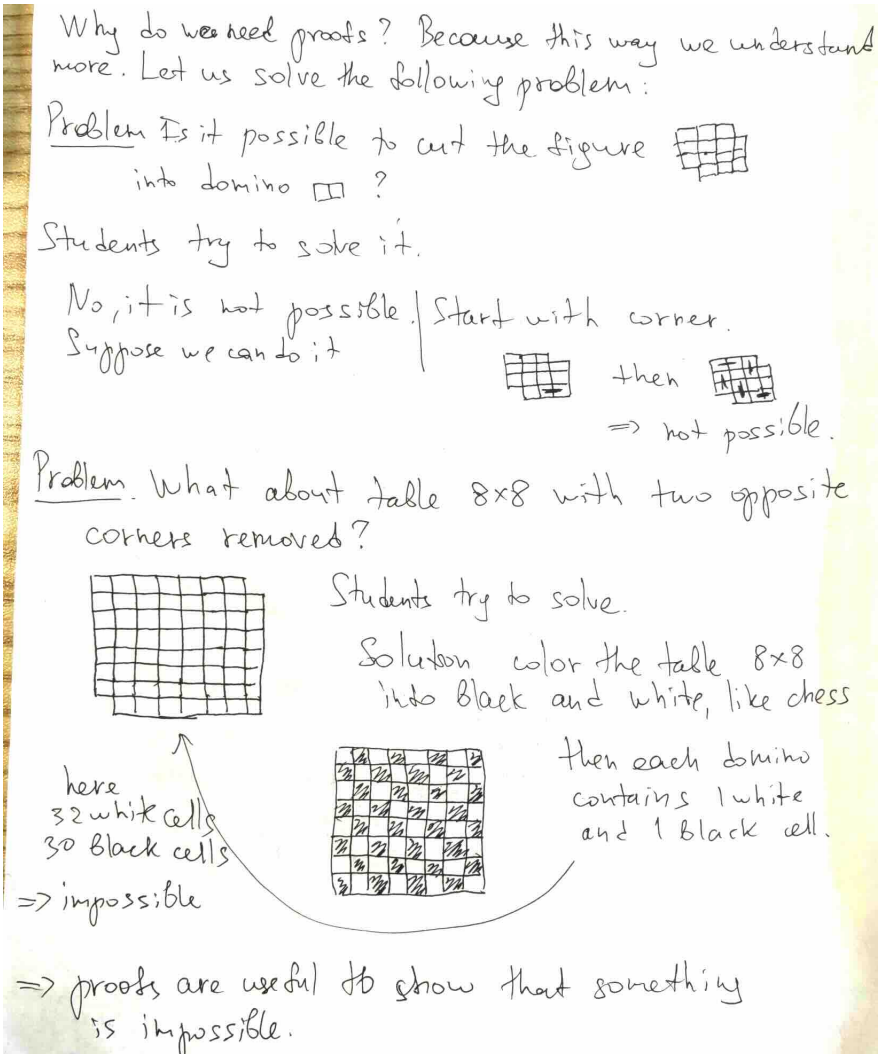
Reflection Logs. It is useful to reflect on how you use AI, so record the tool and its purpose in your diary, and then track how it affects your understanding and your ability to solve problems, write solutions, and read material without any outside help. Note that during the exams or midterms you cannot rely on anything except yourself.

Academic integrity. Copying another person's solution or submitting AI-generated solutions as your own is plagiarism. Violations will be handled under the university's academic integrity policy.

1 Methods of proving

1.1 :: parity, induction

:: :: parity



Many problems become much easier once one notices that some quantity has a fixed *parity* (it is always even or always odd). Once a parity is fixed, any situation in

which that quantity would have the opposite parity is impossible. Sometimes one has to construct this quantity, for example by considering the parity of a sum or product, by pairing objects up, by noticing an alternating pattern, or by colouring objects in two colours.

Example 1. *A grasshopper makes jumps of length 1 m along a straight line and eventually returns to its starting point. Show that it made an even number of jumps.*

Solution. If the grasshopper ends where it started then the number of jumps to the right must equal the number of jumps to the left. Consequently the total number of jumps is even.

Example 2. *Does there exist a closed broken line with seven segments that crosses each of its segments exactly once?*

Solution. Suppose such a broken line existed. Any two crossing segments can be paired. The number of segments must therefore be even, which contradicts the assumption that there are seven segments.

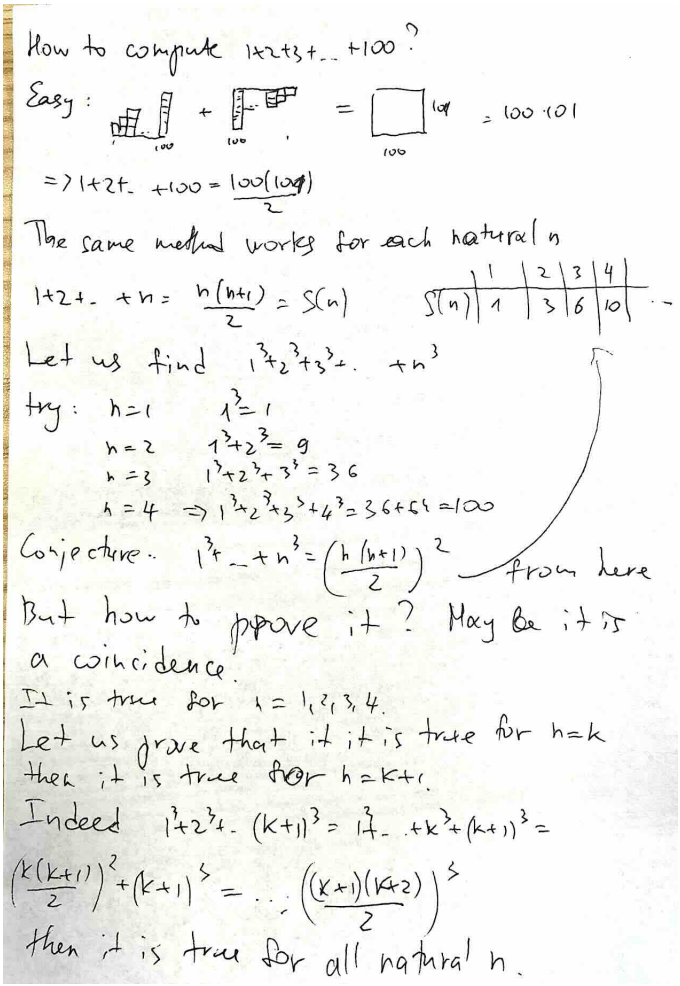
Example 3. *Martians may have any number of arms. One day all Martians joined hands in such a way that no free arms remained. Prove that the number of Martians with an odd number of arms is even.*

Solution. Call Martians with an even number of arms *even* and those with an odd number of arms *odd*. Since the hands form pairs, the total number of hands is even. The total number of hands of the even Martians is clearly even, so the total number of hands of the odd Martians must also be even. But each odd Martian contributes an odd number of hands, so there must be an even number of them.

Such pictures are called *graphs*, they consist of *vertices* (representing Martians) and *edges* (representing a handshaking pair of hands). The number of edges incident to a vertex (i.e. the number of hands of a Martian) is called the *degree* (or *valency*) of a vertex.

We proved that in each graph the number of vertices of odd degree is even.

:: :: induction



The method of mathematical induction is used to prove statements of the form "For every natural number n a certain property holds." Such a statement can be viewed as an infinite chain of assertions: "For $n = 1$ the property holds", "For $n = 2$ the property holds", and so on. The first assertion in the chain is called the base (or the foundation) of the induction and is usually easy to check. One then proves the induction step: "If the assertion with number n is true, then the assertion with number $n + 1$ is true." Sometimes one needs a stronger form of the induction step: "If all assertions with numbers from 1 to n are true, then the assertion with number

$n+1$ is true." There is also the technique of *inductive descent*, in which one proves that if an assertion with number n (with $n > 1$) can be reduced to one or several assertions with smaller numbers and the first assertion is true, then all assertions are true.

If both the base and the induction step have been proved, then all assertions in the chain hold; this is the principle of mathematical induction.

Example 1. *Prove that the number consisting of 243 consecutive ones is divisible by 243.*

Solution. Notice that $243 = 3^5$. We shall prove a more general statement: the number consisting of 3^n consecutive ones is divisible by 3^n. For $n = 1$ the assertion says that 111 is divisible by 3, which is true. Suppose the number consisting of 3^{n-1} ones is divisible by 3^{n-1}. Write

$$\underbrace{11\cdots1}_{3^n \text{ times}} = \underbrace{11\cdots1}_{3^{n-1} \text{ times}} \times \underbrace{10\cdots010\cdots01}_{\text{a block containing only one non zero digit}}.$$

It is not hard to check that the second factor on the right–hand side is divisible by 3. Multiplying a multiple of 3^{n-1} by a multiple of 3 yields a multiple of 3^n. Therefore the number of 3^n ones is divisible by 3^n, completing the induction.

Example 2. *Several lines and circles are drawn in the plane. Prove that the regions into which the plane is divided can be coloured in two colours so that adjacent regions (sharing a segment or an arc) are coloured differently.*

Solution. First erase all the lines and circles, remembering where they were. Colour the entire plane one colour. Then restore the boundaries one by one, recolouring the regions they divide. When adding a line, recolour in the opposite colour all regions on one side of it and leave unchanged those on the other side. When adding a circle, recolour all regions lying inside it and leave unchanged those outside. In this way each time you add a boundary the recoloured regions lie on one side only. Consequently any two neighbouring regions (sharing part of a boundary) always have different colours.

Example 3. *Prove that if $x + \frac{1}{x}$ is an integer, then $x^n + \frac{1}{x^n}$ is an integer for all $n \geq 0$.*

Solution. Set $T_n = x^n + \frac{1}{x^n}$. Note that $T_0 = 2$ and $T_1 = x + 1/x$ are integers. Observe that

$$T_n T_1 = (x^n + \tfrac{1}{x^n})(x + \tfrac{1}{x}) = x^{n+1} + \tfrac{1}{x^{n+1}} + x^{n-1} + \tfrac{1}{x^{n-1}} = T_{n+1} + T_{n-1}.$$

Thus $T_{n+1} = T_n T_1 - T_{n-1}$. By induction on n this recurrence shows that all T_n are integers.

Example 4. (if time permits) *Five robbers have obtained a sack of gold sand. They wish to divide it so that each robber is sure he received at least one fifth of the gold. They have no measuring instruments, but each can judge by eye the amount of a pile of sand. Opinions about the size of the piles may differ. How can they divide the loot?*

Solution (First method). First two robbers divide the sand between themselves: one divides the sack into two piles that he believes equal, and the other chooses his pile. Each of these two divides his share into three equal (to his mind) parts, and the third robber takes one part from each. Now these three each divide their share into four parts and the fourth robber takes one part from each. Finally these four divide their shares into five parts and the fifth robber takes one part from each. Each robber can check that the portion he receives is at least one fifth according to his judgment.

Solution (Second method). Find the most modest robber and give him his portion first. To do so, ask the first robber to separate what he believes to be 1/5 of the sack. Ask the second robber whether the separated part is larger than 1/5: if he thinks it is larger, have him reduce it to what he considers 1/5; if he thinks it is not larger, ask the third robber, and so on. When someone finally agrees that the separated part is exactly 1/5, give that part to the last person who modified it. Among the remaining robbers find the most modest of those and repeat. In the end every robber receives a portion he believes is at least 1/5 of the original amount.